Goals, Requirements, and Constraints for the Design of the TP-6N Field Phone Replacement

last edit: May 19, 2024

authors: Steve Reames, with additions from members of
the NCRC Communications Committee

1.0 Background:

The TP-6N has been the most popular field phone for cave rescue for over 15 years. It has been reliable, lightweight, and does well in the caving environment. There is a need for a replacement field phone that will be designed as a replacement for the TP-6N.

1.1 Who are the Users?

The users of the replacement field phone would be anyone who is involved with communications for cave rescue. This can include ordinary cavers who volunteer their time for teaching or real rescues, search and rescue team members (SAR), or agency personnel such as firefighters, National Park Service (NPS), the Bureau of Land Management (BLM), or United States Forest Service (USFS). This project is generally focused on U.S. users, but can include foreign cave rescue team members.

The NCRC has formed the NCRC Communication Committee (CommComm) to address the needs for supporting cave rescue communications, which includes a replacement for the TP-6N field phone. CommComm has taken the responsibility to design and possibly manufacture this potential replacement field phone. The committee will take into account the needs of the above mentioned users in the design of the field phone.

2.0 Abbreviations and Terms:

POTS: Plain Ordinary Telephone Service. Also referred to as a "landline".

EMC: Electromagnetic Compatibility. Ability of electrical equipment to function acceptably in their electromagnetic environment by limiting the unintentional generation of electromagnetic energy.

NCRC: National Cave Rescue Commission. A division of the NSS dedicated to cave rescue training.

TP-6N: A field phone often referred to as the "NATO phone". Designed in 1973 by the Norwegian Armed Forces.

TA-312: A field phone often referred to as the "crank phone". Designed in the 1950's and deployed in the Korean war.

TA-1: A field phone often referred to as the "sound powered phone". Designed in the 1950's.

RFI: Radio Frequency Interference. An old term, now replaced with EMC.

Opamp: Operational Amplifier. An integrated circuit with many uses in electronics.

SAR: Search and Rescue

PTT: Push To Talk. A switch that enables the microphone of a field phone.

3.0 Nomenclature:

A number surrounded by brackets, such as [1], indicates the priority of the requirement. There are three categories:

[1] -- Required: Cannot be changed without revisiting the purpose of the entire project.

[2] -- Strong Want: We are willing to put extra effort to accomplish this requirement.

[3] -- Would Be Nice: An extra feature that would make the product easier to use.

4.0 Requirements:

The requirements in this section are a combination of military specifications and real-world use of the field phones. The published military specs are very sparse and are completely inadequate for designing a replacement field phone (see Appendixes). When military specs are not available we fall back to measurements of physical units, knowing that we will be working with typical values rather than worst case values.

Section 4.1

[1] Backward compatible with the existing 2-wire military field phones, specifically the TP-6N ("NATO" phone), TA-312 ("Crank phone"), and the TA-1 (Sound-powered phone).

Description: Backward compatible means that the new phone can be added to an existing phone system with no adverse effect. The physical implementation can be different, for example the TA-312 uses D-cell batteries and the new proposed phone will use AA batteries. But the main functions of duplex voice communication with high-voltage ringing signal must be maintained.

Section 4.2

[1] The replacement unit will work with the standard 2-wire communication system. It will be designed to use lines with no 48V bias (LB, not CB, in military speak). It may work with lines DC voltage applied but is not required. Wires for cave rescue are assumed to have no DC bias.

Description: Conventional POTS telephones use a central 48V battery for powering the phone and for signaling. This is not required for cave rescue applications and the design of inductors that need to support 48V DC lines is unnecessarily complex. There is also the question of

range. Two TP-6N units can communicate up to 14 miles (wet, WD-1 wire). The TA-312 also can go 14 miles (wet, WD-1 wire). The TA-1 is rated at 4 miles, and the EE-8 is rated at 7 miles. It is also known that each additional attached phone reduces the range. The proposed unit has not been tested for range. The standard WD-1 military wire is assumed, but the system should work with any pair of wires, though the range may be reduced.

Section 4.3

[2] Power: The replacement unit should operate from 3.0V to 5.0V. The quiescent current should be equal to or less than 7.5 mA. Typical power would be three AA batteries (4.5V to 3.0V) or a single 18650 Lithium-ion battery (4.2V to 3.0V).

Description: This is based on the typical use of three AA batteries for the TP-6N. The TP-6N was designed to use three D-cells, but the military wanted the unit to run for weeks. The capacity of an alkaline AA is approximately 2300 mAh (for a 1.0V dropout). So the quiescent run time would be 300 hours. For a Sanyo 3400mAh lithium-ion battery the run time would be 450 hours. The actual run time will be less due to use of the microphone and the ringer. It is rare for a cave rescue to run for 4 days (100 hours). In that situation it is reasonable to expect that someone would bring replacement batteries if the rescue ran that long. So a quiescent current of 7.5 mA is very conservative and can be increased if needed. The maximum permitted voltage of 5.0V is because many of the low-power opamps have a maximum voltage of 5.5V.

Section 4.4

Ringing Voltage: Ringing is a high-voltage signal used to alert other users on the line. The function of ringing is a type [1] requirement, but the exact voltage and frequency is a type [2] requirement. The ringing is expected to be an AC signal in the 40 to 100 Vpeak range. The ringer circuitry must not be damaged by a short circuit on the line.

Description: The generation of the ringing signal is one of the hardest design challenges for the replacement field phone. The circuit must work at the minimum input voltage (3.0V) but generate a 60 V (typical) signal that is switched at a 20 Hz (typical) to create an 120 Vp-p waveform on the line. The TP-6N has an amazing low part count for their circuit but required custom magnetics that we do not have access to nor the construction of said magnetics. We, being small-shop designers and manufacturers, are limited by off-the-shelf components or building components (such as magnetics) that are easily assembled with technician skills and do not require special tools.

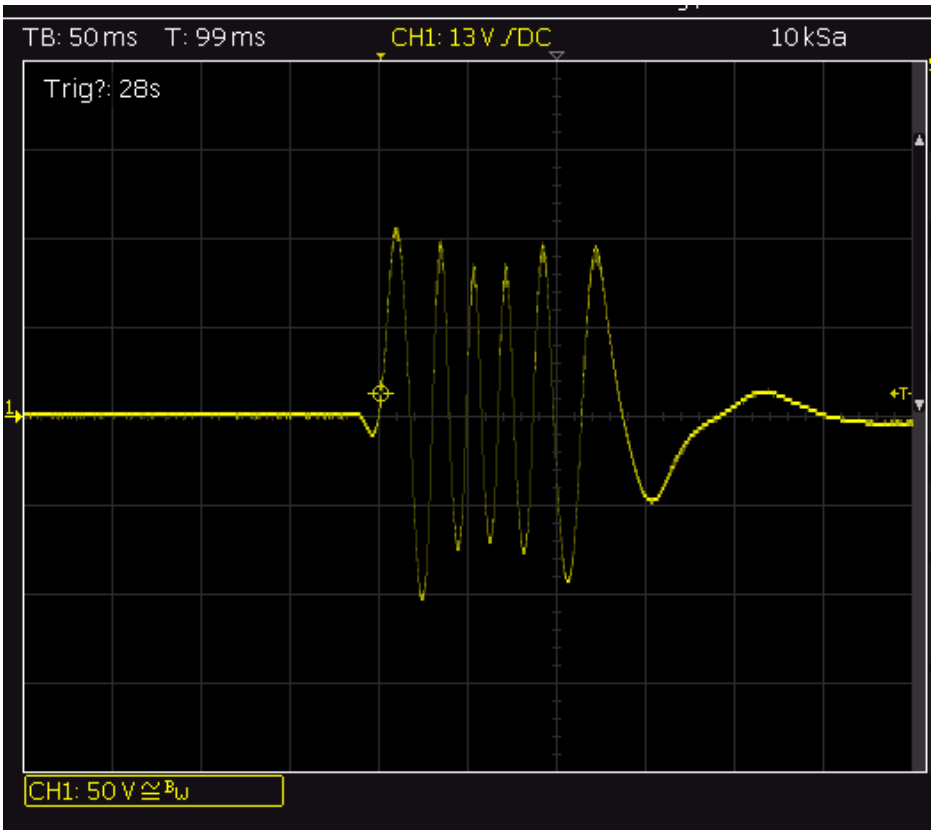
The TA-1 uses a hand generator to create the ringing signal. One squeeze on the trigger creates the output voltage. The TA-312 has a crank generator on the side of the unit to create the ringing signal. On the receiving end, the TA-312 has a hammer on a diaphragm that creates a drumming sound. The specs for the TA-312 say the ringing signal is 20 Hz at 90 to 100 V (but does not indicate if that voltage is Vpeak, Vp-p, or RMS).

The TP-6N uses a solid-state ringing signal. When the button is pressed the circuit creates a 20 Hz or 25 Hz signal (depending on whether the unit is Norwegian or Swedish). The specification is that with a 4.5 V battery voltage the output is 60 Vrms; with a 3.0 V battery the output voltage

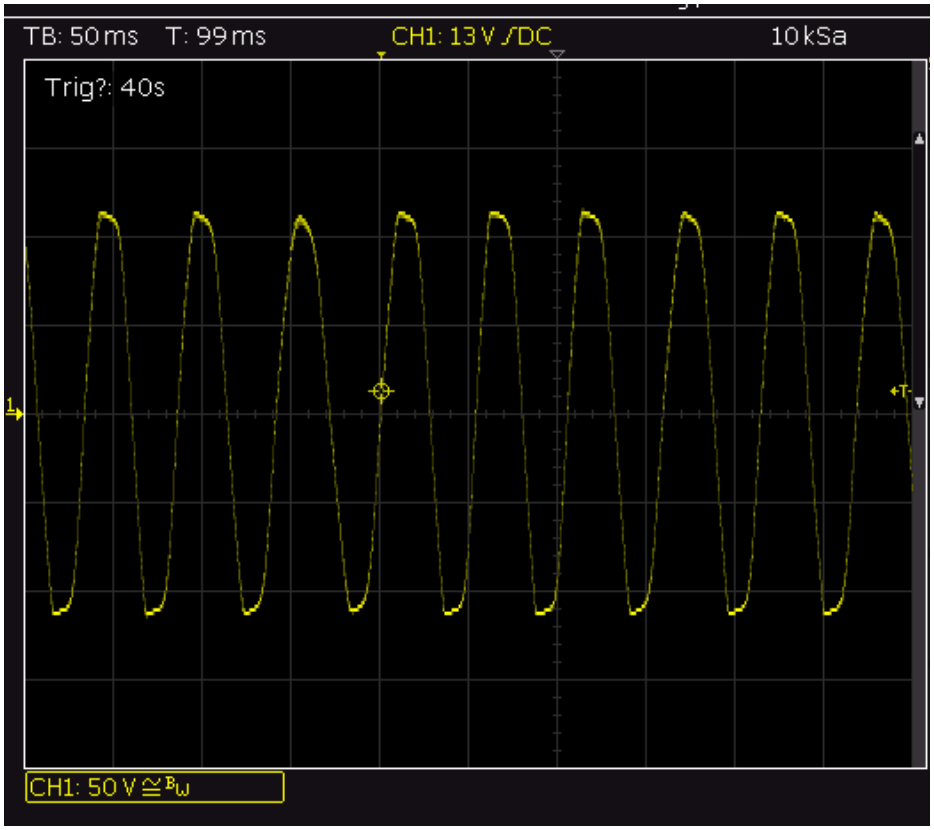
is 38 Vrms (with a 3 Kohm load). It is known that on long lines the TP-6N does not have enough energy to trigger the sounder of a TA-312.

Commercial ringing circuits (used with local PBX systems) often have specifications of an output of voltage of 70 to 91 Vrms with an input power of 2 to 3 Watts. (Though available on eBay, they all require an input voltage of 12V.)

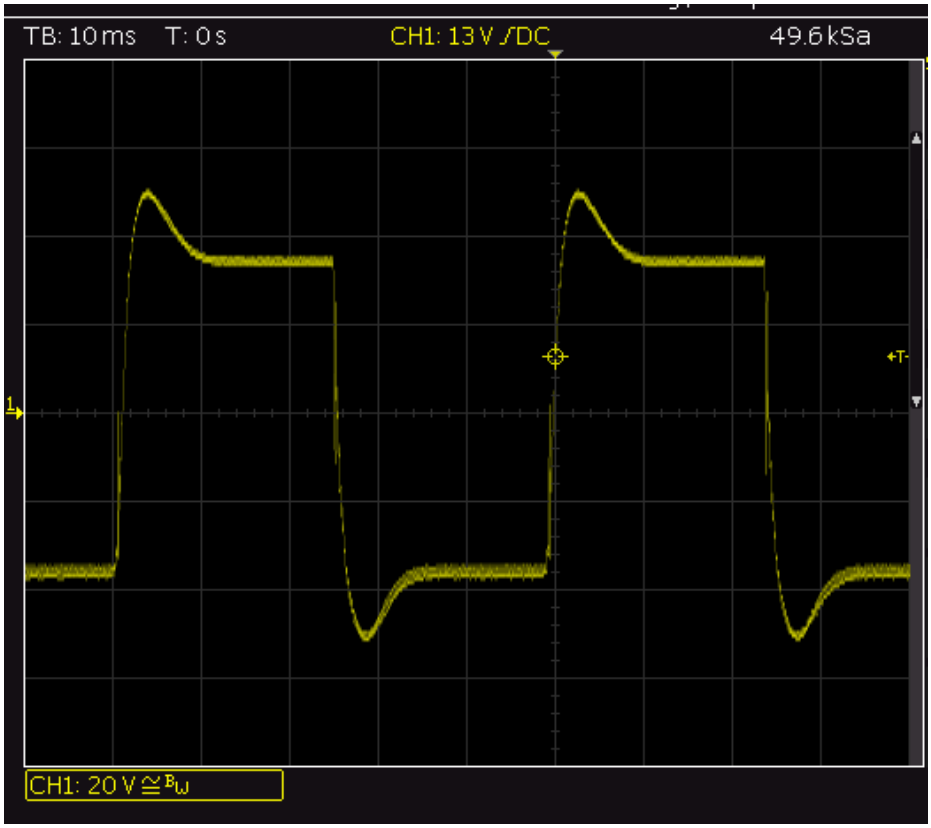
The telephone system (POTS) specifies the impedance of one REN (Ringer Equivalent Number) to be approximately 7 Kohm (or more) at 20 Hz. This is consistent with our measurement of 35 Kohm at the input of a TP-6N. Though REN is specified for POTS, the military phone system is derived from the POTS.



Ringing signal for the TA-1 (with 500 ft line and TP-6N load)



Ringing signal from a TA-312 (with 500 ft line and TP-6N load)



Ringing signal from TP-6N (with a 500 ft line and a TA-312 load)

Section 4.5

[2] Incoming Ringing Signal (audio): An incoming ringing signal will be heard in the earpiece of the handset. No additional circuitry is needed to receive the signal.

Description: The current design uses the same system as the TP-6N; the clicking from the ringing signal is heard in the earpiece. This is different from almost all military field phones; they typically have a separate sounder in the base unit. The TA-312 has a "drum" that vibrates at the ringing signal frequency. The British PTC-414 (circa 1993) has a piezoelectric buzzer that oscillates at 2 KHz, modulated by the 20 Hz ringing signal. This information is just for reference in the case we wish to change how the incoming ringing signal is indicated.

Section 4.6

[2] Incoming Ringing Indicator (visual): When a ringing signal is present, whether it is being generated in the unit or being received from a remote unit, a red LED flashes (at 20 Hz). An additional troubleshooting feature [3] of the TP-6N is that if the unit is open circuited (or not connected to the line) then the user can hear the ringing in the earpiece, but no LED is present. If the unit is shorted then the LED is visible, but no ringing is heard in the earpiece.

Description: The LED in the TP-6N appears to be type 1-3/4 (5 mm) which makes sense because red LEDs were just entering the market in the early 1970's. No other field phone (of that vintage) has a visible indicator. But it has been found to be very handy, particularly for troubleshooting. The LED on the TP-6N is very hard to see in daylight but works very well in cave conditions. These days there are much more efficient LEDs that only cost a few more pennies.

Section 4.7

[1] Weight and Volume: The weight and volume should be equal or less than a TP-6N.

Description: The weight of the TP-6N handset and cord is 0.73 lbs, the base unit of the TP-6N is 2.16 lb, and the entire package, including the pouch and D-cell batteries is 3.64 lbs. This is an unfair comparison because we never use D-cells, instead we use the insert with three AA batteries. Subtracting the difference between the D-cells and the AA cells, we should be looking for a target weight of 2.97 lbs. Volume is also a key parameter for cave rescuers. The existing TP-6N (total package) is approximately 10.2" x 7.7" x 2.6". A smaller package would enable a rescuer to move faster, or carry more gear at the same speed. Both are valuable attributes.

The intent of the project is to leverage techniques and processes that did not exist or were too expensive to implement in the 1970's. Some examples are 3D printing, inexpensive PCBs, and surface-mount devices (SMDs).

Section 4.8

[1] Full Duplex Communication: The existing military phone system uses full duplex communications. Specifically, when talker A is speaking, they still can hear speech from person B. This requires a "hybrid" circuit which combines 4-wire communication (microphone and earpiece) into a 2-wire system which is attached to the field wire. This is how POTS works.

Description: The addition of a hybrid circuit is an additional complexity which is generally avoided. But to be compatible with existing field phones then it is required. The other approach, a half duplex system, is what is used with radios -- when the talker keys the microphone then no one else can interrupt the transmission. The use of a hybrid gives the ability of the receiver to give instant feedback to the transmitter (e.g. "hold on, give me that number slower", or "break, break, priority message incoming"). One could argue the utility of that feature, but since we want to be compatible with existing field phones then the decision is done. There is the question of why do we have a PTT button on our field phones? POTS does not have it. The answer is to keep the noise down on the wires. Only the person who has keyed their PTT will enable their microphone.

Section 4.9

[1] Electromagnetic Compatibility: Any electronic circuit has the potential to radiate electromagnetic energy and interfere with nearby communications. The Federal Communications Commission (FCC) is responsible for regulating the frequency and intensity of any EM emissions. There are three options for handing the FCC regulations. 1) Ensure by design that any potential radiation will be below any of the frequencies that are regulated by the

FCC. 2) Submit sample circuits to a test lab for certification that will confirm that the circuit is compliant. 3) Ignore the FCC regulations because it is unlikely we would be caught. We must decide which of the three options we choose to follow.

Description: Reading the regulations for EMC can be overwhelming. But in the U.S. battery operated devices are exempt as long as there are no clocks higher than 1.705 MHz. So the current Blackbird project is exempt. There is a caveat is that the product is not exempt if the product is running and connected (directly or indirectly) to an AC power line. So the field phone cannot be running while plugged into a battery charger. One of the proposed features is the ability to run off external batteries, such as 12V car battery. This would probably require a switching power supply to convert the 12V into 4.5V for the field phone circuitry. Most switching power supplies (at these powers and voltages) switch at 150 KHz to 1 MHz, so we are still covered. (see Appendix D.)

Technically, there is a fourth option: do it yourself kits. For many years it has been assumed that if you build a kit then it is exempt. Though this may have been true in the past, it is not correct anymore. But it is not clear who is responsible for compliance -- the kit manufacturer or the user. A dated link from Sparkfun is interesting reading:
https://www.sparkfun.com/tutorials/398

It would be nice to have the field phone covered in other countries, specially the EU. That would be a type [3] requirement. In recent years the EU has been regulating lower frequency emissions. In the past you could be safe with any frequency below 30 KHz. Recent changes apply to some circuits as low as 9 KHz (think of buried wires for guiding a robot lawn mower). Since certifications generally cost $2500 and up, it is not cost efficient for us to comply with EU regulations. But if we do have to do testing for the U.S. then it typically only costs another 10% to certify for the EU also.

EMC certification is a complex topic and cannot be easily summarized in these pages.

5.0 Requirements for Future Text-Based Communications System

(To be determined)

Notes from Andy Sheaff in April 2024:

Digital Text System Constraints (no particular order):

* Use WD-1 field wire for an communications because teams already have this wire and are unlikely to change.

* Avoid baseband digital data to keep Comms Operators from having to listen to the digital data; Parallel voice and data may be possible.

* Operate without termination because this just happens.

* Operate with or without stubs and branches of arbitrary length because the location of field phones may be in different and down forked passages. Field wire goes where it needs to.

* Operate with 10 km of wire between stations from Steve's recollection of the most distance rescue.

(Note: Steve's recollection was from the Fisher Ridge rescue reported by Tim Miller. They used 9,000 feet of wire, which is different than 10 km, though 10 km would be good goal - Steve R.)

* Devices should not be damaged by a short on the field wire because shorts do occur.

* Operate with a variable number of field phones attached as the number of field phones varies.

Appendix A:

Technical Data for the TP-6N

TECHNICAL DATA

Description	Measuring condition	Nominal value
Frequency range		300-3400 Hz
Impedance	300-3400 Hz speech mode 300-3400 Hz not activated «Press-to-talk» 300-3400 Hz not activated «Press-to-talk and listen» (1 kHz)	600 ohm
Speech transmission	Measured with a sound pressure of 94 dB relative 20×10^{-6} Pascal. 1000 Hz and 600 ohm load	387 mV
Speech reception	Sensitivity with a generator voltage of 570 mV in series with 600 ohm, 1000 Hz and a 6 cm^3 acoustical coupling.	103 dB relative 20×10^{-6} Pascal
Call frequency		25 Hz
Call voltage	At 3 kohm load, 4.5 V battery voltage in series with 1.5 ohm. At 3 kohm load, 3.0 V battery voltage in series with 1.5 ohm. Voltage measured with mean value indicating instrument (RMS)	60 V 38 V
Current consumption	Speech mode with battery voltage of 4.5 V. Calling with battery voltage of 4.5 V. 1.5 ohm in series with voltage source.	7.5 mA 430 mA

Description	Measuring condition	Nominal value
Battery voltage	(3xBA30, Lithium or eqv.)	4.5 V
Acoustical warning	Measured in a test chamber with reference to a special test layout with a 20 V/25 Hz incoming call signal.	62 dBA relative 20×10^{-5} N/m^2
Hybrid balance	Attenuation at 1000 Hz and 600 ohm termination, 0 km line.	18 dB
Insulating resistance	With test voltage less than 40 V DC.	250 Mohm
Interference protection		Satisfied
EMP protection	NATO approved (March 1979)	
Temperature		−40°C +55°C

	Dimensions			Weight
	Length	Width	Height	
Handset w. cable	210 mm	55 mm	70 mm	0.33 kg
Instrument ex. batt.	233 mm	100 mm	47 mm	0.98 kg
Pouch	260 mm	195 mm	65 mm	0.33 kg
Complete Set with Batteries and bag	*)Depending on batt.type			1.65 kg*

We reserve the right to change the specifications without prior notice.

Appendix B:

Specifications for the TA-312/PT

Section II. DESCRIPTION AND DATA

1-7. Purpose and Use

The TA-312/PT is a two wire, battery-operated field telephone. It may be used in a simple point-to-point voice frequency (vf) wire communication link or in any two-wire ringdown subscriber position of a telephone communications system.

1-8. Technical Characteristics

a. Voltage requirements . . . . . 3 volts dc.

b. Frequency range . . . . . 300 to 3,200 Hz.

c. Signaling:
Frequency . . . . . 20 Hz.
Voltage . . . . . 90 to 100 volts.

d. Station-to-station distance.

Wire description and type or gauge	Maximum distance (miles)	
	Wet	Dry
Wire, Cable, Telephone WD-1/TT (field wire).	14	22
Wire, Cable, Telephone WD-14/TT (field wire).	14	22
	Loaded	Not Loaded
Cable Assembly, Telephone CX-1065/G (spiral-four)	29	50
Cable Assembly, Telephone CX-162/G (five-pair cable).		23
Lead-covered cable, 19-gauge		30
Lead-covered cable, 22-gauge		20
Lead-covered cable, 24-gauge		
Open-wire lines, W-2 #l 4 AWG copper.		230
Open-wire lines, W-74 #l 2 AWG copper.		520

Appendix C:

Specifications for the TA-1/PT

1-10. EQUIPMENT DATA

- VOICE TRANSMISSION RANGE . . . . . 4 miles using field wire WD-1/TT (approximate)
- SIGNAL RANGE . . . . . 4 miles using field wire WD-1/TT (approximate)
- FREQUENCY RANGE . . . . . 300 to 4000 Hz
- SIGNAL VOLTAGE . . . . . 65 to 80 Vac at 20 Hz
- OPERATING TEMPERATURE . . . . . –40 to +131°F

Quantity	Item	Dimensions (in.) Height	Depth	Width	Unit weight
1	Handset, including 1 Earphone Element TA-118/PT 1 Microphone Element TA-121/PT	(less cord)	4	3 1/4	2 3/4 lb
1	Telephone set case (rubberized) with carrying strap (Order no. 24977-Phila-55, Serial nos. 1 through 9300)	10	4 1/4	3 1/2	14 oz.
1	Telephone set case (plastic) with carrying strap (Order no. 24977-Phila-55, Serial nos. 9301 and higher. Order nos. 38695-Phila-56 and 21101-Phila-57)	9 5/8	3 9/16	5 1/8	11 1/2 oz
1	Telephone set case (plastic) with carrying strap (Order no 19629-Phila-58)	10	4 1/4	3 1/2	9 oz.

Appendix D:
FCC Title 47, Chapter 1, Subchapter A, Part 15, Subpart B:

§ 15.103 Exempted devices.

Except as provided in paragraph (j) of this section, the following devices are subject only to the general conditions of operation in §§ 15.5 and 15.29, and are exempt from the specific technical standards and other requirements contained in this part. The operator of the exempted device shall be required to stop operating the device upon a finding by the Commission or its representative that the device is causing harmful interference. Operation shall not resume until the condition causing the harmful interference has been corrected. Although not mandatory, it is strongly recommended that the manufacturer of an exempted device endeavor to have the device meet the specific technical standards in this part.

(a) A digital device utilized exclusively in any transportation vehicle including motor vehicles and aircraft.

(b) A digital device used exclusively as an electronic control or power system utilized by a public utility or in an industrial plant. The term *public utility* includes equipment only to the extent that it is in a dedicated building or large room owned or leased by the utility and does not extend to equipment installed in a subscriber's facility.

(c) A digital device used exclusively as industrial, commercial, or medical test equipment.

(d) A digital device utilized exclusively in an appliance, e.g., microwave oven, dishwasher, clothes dryer, air conditioner (central or window), etc.

(e) Specialized medical digital devices (generally used at the direction of or under the supervision of a licensed health care practitioner) whether used in a patient's home or a health care facility. Non-specialized medical devices, *i.e.*, devices marketed through retail channels for use by the general public, are not exempted. This exemption also does not apply to digital devices used for record keeping or any purpose not directly connected with medical treatment.

(f) Digital devices that have a power consumption not exceeding 6 nW.

(g) Joystick controllers or similar devices, such as a mouse, used with digital devices but which contain only non-digital circuitry or a simple circuit to convert the signal to the format required (e.g., an integrated circuit for analog to digital conversion) are viewed as passive add-on devices, not themselves directly subject to the technical standards or the equipment authorization requirements.

(h) Digital devices in which both the highest frequency generated and the highest frequency used are less than 1.705 MHz and which do not operate from the AC power lines or contain

provisions for operation while connected to the AC power lines. Digital devices that include, or make provision for the use of, battery eliminators, AC adaptors or battery chargers which permit operation while charging or that connect to the AC power lines indirectly, obtaining their power through another device which is connected to the AC power lines, do not fall under this exemption.

(i) Responsible parties should note that equipment containing more than one device is not exempt from the technical standards in this part unless all of the devices in the equipment meet the criteria for exemption. If only one of the included devices qualifies for exemption, the remainder of the equipment must comply with any applicable regulations. If a device performs more than one function and all of those functions do not meet the criteria for exemption, the device does not qualify for inclusion under the exemptions.

(j) Notwithstanding other provisions of this section, the rules governing certification apply to any equipment produced by any entity identified on the Covered List, as established pursuant to § 1.50002 of this chapter, as producing covered communications equipment.